

Problem Set 4

Kevin Ruiz

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1 Question 1

- a The first regression implies that age is a determinant in explaining wage, while the second one would tell us the opposite. The relationship between how wage could determine age. I would say the first is more consistent with economic theory since we keep in mind that the relationship between wage and age is not that a higher wage implies that we magically become older but instead that as we get older our wage increases due to experience in the labor market.

- b i The first one is true since as we can see $\frac{1}{.1172468} \neq .0110603$. This is mathematically done by,

$$\frac{wage_i - \alpha_i - \epsilon_i}{\beta_1} = age_i$$

However, if we look at what age is equal to

$$age_i = \alpha_2 + \beta_2 wage_i + \epsilon_i$$

Thus, for this to hold we would need the intercepts equal to zero as well as the error term. Which is an unreasonable assumption.

- ii Since RSS is dependent on variance of its predictor then we can notice the following two formulas.

$$RSS_{wage} = (1 - r^2)var(wage)$$

$$RSS_{age} = (1 - r^2)var(age)$$

Thus since both of these depend on different variances then they are clearly not going to be equal unless, the variances are equal. The rest, follow this idea for TSS and ESS.

For ESS

$$\hat{y}_{wage} \neq \hat{y}_{age}$$

thus when calculating it on the average input for it's own regression it will not equal each other. Especially since they are based on different errors thus they cant be equal.

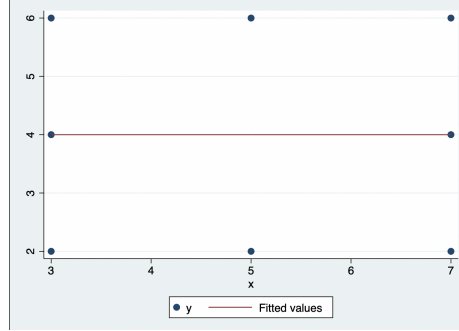


Figure 1: Fitted Reg for Data Set 1

- iii Since R^2 is calculated as the proportion of variance explained, and the numerator involves the square of the covariance $\sum_i (x_i - \bar{x})(y_i - \bar{y})$, swapping x and y does not change this value. Mathematically, we have

$$R_{wage}^2 = \frac{\hat{\beta}_1^2 \sum_i (age_i - \bar{age})^2}{\sum_i (wage_i - \bar{wage})^2} = \frac{[\sum_i (age_i - \bar{age})(wage_i - \bar{wage})]^2}{\sum_i (age_i - \bar{age})^2 \sum_i (wage_i - \bar{wage})^2} = R_{age}^2$$

Thus, the R^2 values are equal for both regressions.

- iv The t -statistics for the slope coefficients are calculated by

$$t_{\hat{\beta}_1} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}, \quad t_{\hat{\beta}_2} = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2)}$$

Since in simple regression the t -statistic is related to R^2 by

$$t^2 = F = \frac{R^2}{1 - R^2} \cdot (n - 2),$$

and we already showed that $R_{wage}^2 = R_{age}^2$, it follows that

$$t_{\hat{\beta}_1} = t_{\hat{\beta}_2}.$$

Thus, even though the slopes are different, the t -statistics are equal.

- c That a big part of finding causal relationship is really understanding that x is the one that affects y and not the other way around. That the issue might be of reverse causality because we wouldn't say that your wage affects your age but instead that age is the true determinant of wage.

2 Question 2

We see that the R^2 for this data set is 0. As we can see in figure 1 we end up seeing that the fitted values vs our y and x where we have the same amount of points above our regression line as below.

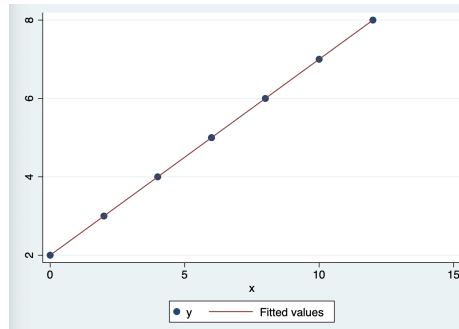


Figure 2: Fitted Reg for Data Set 2

3 Question 3

We see that the R^2 for this data set is 1. As we can see in figure 2 it is because our x and y follow a linear relation.

4 Question 4

Yes, mathematically, if we just add an infinite amount of statistically important variables, we would still find a higher R^2 . However, from the example from class we could also have an issue that some of the variables together explain a big part of our variable of interest, however, if they also explain the same parts then our regression would fit them as insignificant even though they help explain the relationship.

5 Question 5

- a The first intercept α_1 is the average for males, while β_1 ends up being the difference between the average for males vs females. Now for the second intercept α_2 is the average for female where β_2 is the difference in the averages with female as the base group and male.
- b The relationship ends up being that they get the same magnitude but opposite signs.

$$\beta_1 = \bar{w}_f - \bar{w}_m$$

$$\beta_2 = \bar{w}_m - \bar{w}_f$$

Thus we can see that it ends up being the same subtraction however flipped.